

ATHMEEYA M KASHYAP

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EDUCATION

IIIT Hyderabad

2024 - 2029 (Expected)

B.Tech Computer Science + MS Computational Natural Sciences

Hyderabad, India

- Coursework: Data Structures & Algorithms, Discrete Mathematics, Linear Algebra, Probability & Random Processes, Design & Analysis of Software Systems, Machine & Data Learning.

PROFESSIONAL EXPERIENCE

Software Engineer Intern

Jan 2026 - Apr 2026

SVP India (Social Venture Partners India)

Hyderabad, India

- Designed a normalized PostgreSQL schema across **12 relational tables** for SVP India's Partner Engagement Dashboard using Prisma ORM and Express.js.
- Developed and secured **60+ REST API endpoints** with JWT/bcrypt authentication and role-based access control using React.js, TypeScript, and Express.js.
- Automated bulk partner data ingestion through custom seed scripts, eliminating **100%** of manual data entry across partner lifecycle records.

PROJECTS

Felicity Fest Management System 🌀 | React 18, Node.js, Express.js, MongoDB, Socket.IO

Jan 2026 - Mar 2026

- Architected a **3-role MERN** event management platform with **70+ REST endpoints** and **9 MongoDB models** using Mongoose discriminator-based role inheritance; integrated JWT/bcrypt authentication, RBAC-protected routes.
- Built **2 real-time communication systems** using Socket.IO with room-isolated discussions, threaded replies, moderation tools and message history; added **QR attendance scanning** via html5-qrcode, eliminating native app dependency.
- Designed hackathon team registration with cryptographic **8-character** invite codes, merchandise approval workflows, atomic MongoDB stock updates, and organizer CSV export; integrated Multer uploads with a **10MB** type-filtered limit.

YouMatter Wellness Platform 🌀 | React 19, Express.js, Python, Flask, Prophet, TensorFlow.js

Oct 2025 - Nov 2025

- Ranked 9th out of 100+ teams** at the Star Union Dai-ichi National Tech Hackathon; Implemented a **3-tier** gamified wellness platform modeling **5 wellness dimensions** as a **6-node** interconnection graph with a **30%** ripple-effect engine.
- Devised a dual ML forecasting pipeline with Facebook Prophet using a **90-day** synthetic training window and **7-day** prediction horizon, backed by a TensorFlow.js fallback to keep analytics functional during backend ML service outages.
- Engineered a centralized **5-action useReducer** state machine and decoupled **4 independent page components** via a native CustomEvent bus, eliminating prop drilling across **10+ nested components** while rendering **4 chart types**.

RESEARCH EXPERIENCE

Undergraduate Researcher, CCNSB Lab, IIIT Hyderabad

Jan 2026 - Present

Advisor: Prof. Chittaranjan Hens | Complex Networks, Dynamical Systems & Phase Transitions

Hyderabad, India

Flash Crash Dynamics in Networked Oscillators 🌀 | Python, C++17, SciPy

- Modeled synchronization dynamics using Kuramoto and Stuart-Landau oscillators on ER and BA networks across **525 simulations** with system sizes ranging from $N = 50$ to $N = 800$.
- Optimized the ODE integrator in C++17 using RK4, achieving a **30x speedup** over Python and scaling execution for large network configurations.
- Simulated abrupt **88%** liquidity withdrawal as a coupling-drop flash-crash event and recovered $K_c = \sqrt{8/\pi} \approx 1.5957$ within **2%** using finite-size scaling and synchronization-based early-warning signals.

ACHIEVEMENTS

INMOTC Selection (2022): Selected for INMO Training Camp at ICTS-TIFR, Bengaluru (**1 of 22 from Karnataka**).

INMO Qualifier (2022): Qualified for the Indian National Mathematical Olympiad (**among the top ~300 nationally**).

NSEA, Top 1% Karnataka: Top 1% statewide in the National Standard Examination in Astronomy (IAPT).

Karnataka SSLC State Topper: Scored **625/625** in the Karnataka State Board Examination.

SGFI National Swimming: Represented Karnataka in Swimming at the School Games Federation of India Nationals..

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, TypeScript, SQL, Bash/Shell

Frameworks & Libraries: React.js, Node.js, Express.js, Flask, Prisma ORM, Mongoose, Socket.IO, Tailwind CSS, NumPy, SciPy, Pandas, Scikit-learn, NetworkX

Databases & Tools: PostgreSQL, MongoDB, SQLite, Git, Docker, AWS, Linux

Core Competencies: Data Structures & Algorithms, System Design, REST API Design, OOP, Graph Theory, Probability Theory